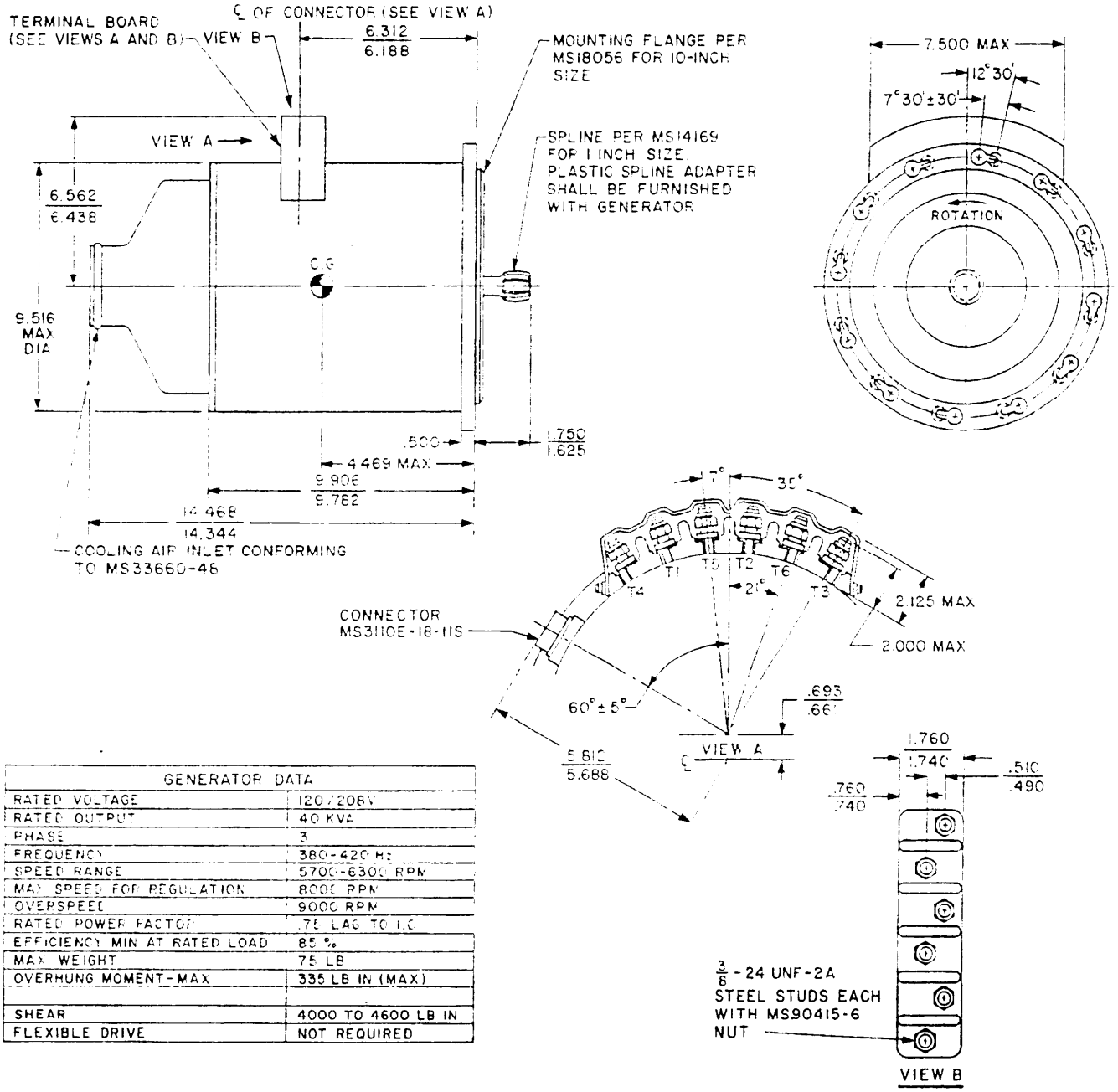
MILITARY SPECIFICATION SHEET

GENERATOR SYSTEM, 40 KVA, 400 HERTZ
ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS TYPE

THIS SPECIFICATION HAS BEEN APPROVED BY THE NAVAL AIR SYSTEMS COMMAND, DEPARTMENT OF THE NAVY, AND IS AVAILABLE FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

THE COMPLETE REQUIREMENTS FOR PROCURING THE GENERATOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF MIL-G-21480.

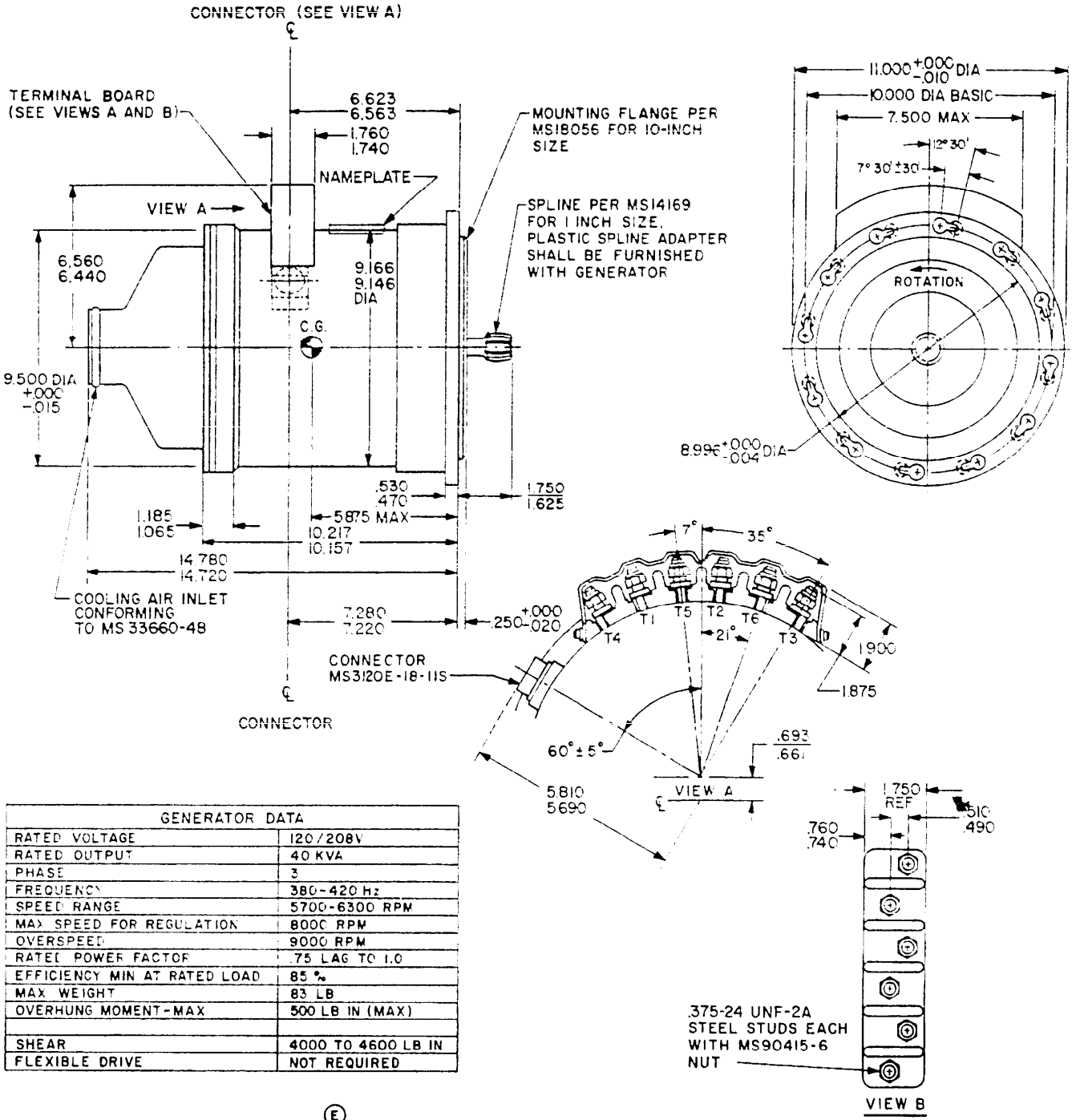


GENERATOR DATA	
RATED VOLTAGE	120/208V
RATED OUTPUT	40 KVA
PHASE	3
FREQUENCY	380-420 Hz
SPEED RANGE	5700-6300 RPM
MAX SPEED FOR REGULATION	8000 RPM
OVERSPEED	9000 RPM
RATED POWER FACTOR	.75 LAG TO 1.0
EFFICIENCY MIN AT RATED LOAD	85 %
MAX WEIGHT	75 LB
OVERHUNG MOMENT-MAX	335 LB IN (MAX)
SHEAR	4000 TO 4600 LB IN
FLEXIBLE DRIVE	NOT REQUIRED

FIGURE 1. M21480/12-1, GENERATOR

FSC 6115

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GENERATOR DATA	
RATED VOLTAGE	120/208V
RATED OUTPUT	40 KVA
PHASE	3
FREQUENCY	380-420 Hz
SPEED RANGE	5700-6300 RPM
MAX SPEED FOR REGULATION	8000 RPM
OVERSPEED	9000 RPM
RATED POWER FACTOR	.75 LAG TO 1.0
EFFICIENCY MIN AT RATED LOAD	85 %
MAX WEIGHT	83 LB
OVERHUNG MOMENT-MAX	500 LB IN (MAX)
SHEAR	4000 TO 4600 LB IN
FLEXIBLE DRIVE	NOT REQUIRED

(E) FIGURE 2. M21480/12-2, GENERATOR

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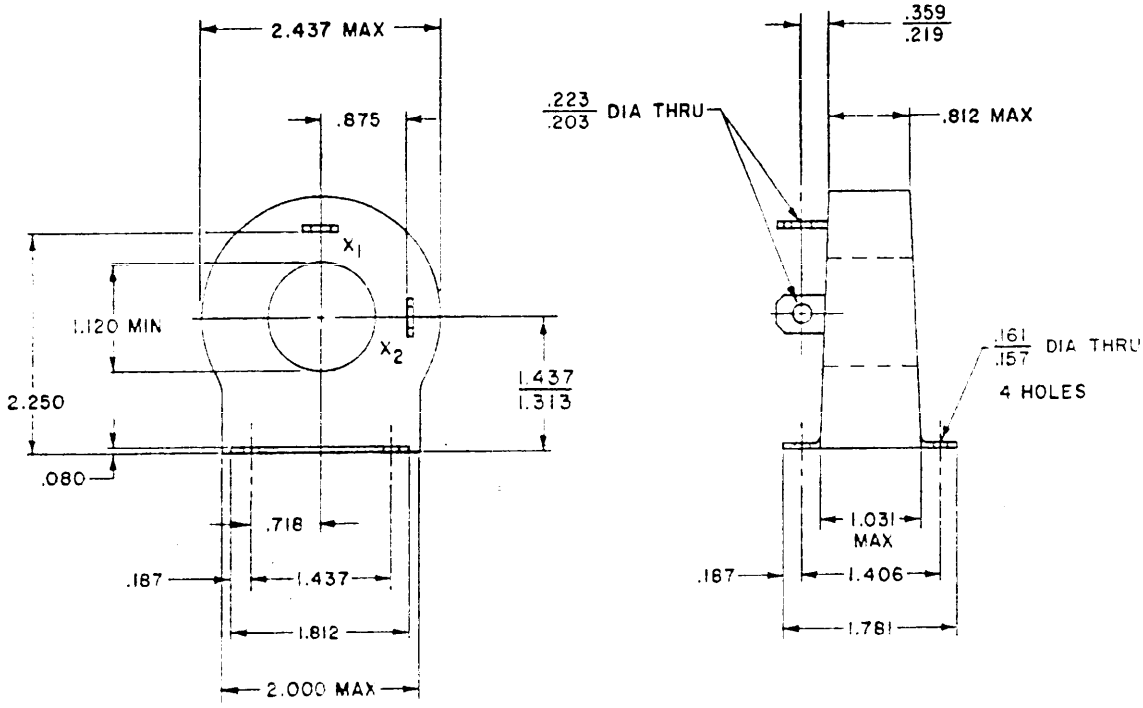


FIGURE 3
Ⓔ LOAD END CURRENT TRANSFORMER, WT. 0.4 LB. MAX
PN M21480/12-5

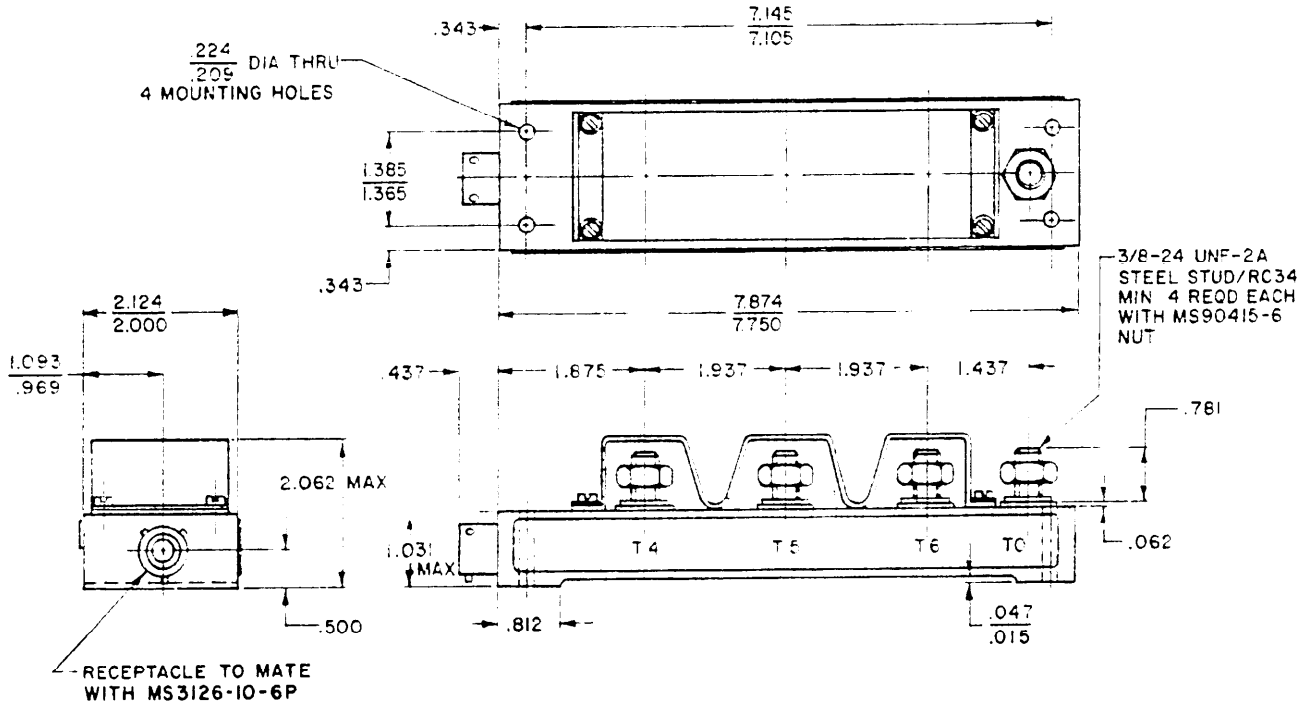


FIGURE 4
Ⓔ GROUND END CURRENT TRANSFORMER, WT. 2.0 LBS. MAX
PN M21480/12-6

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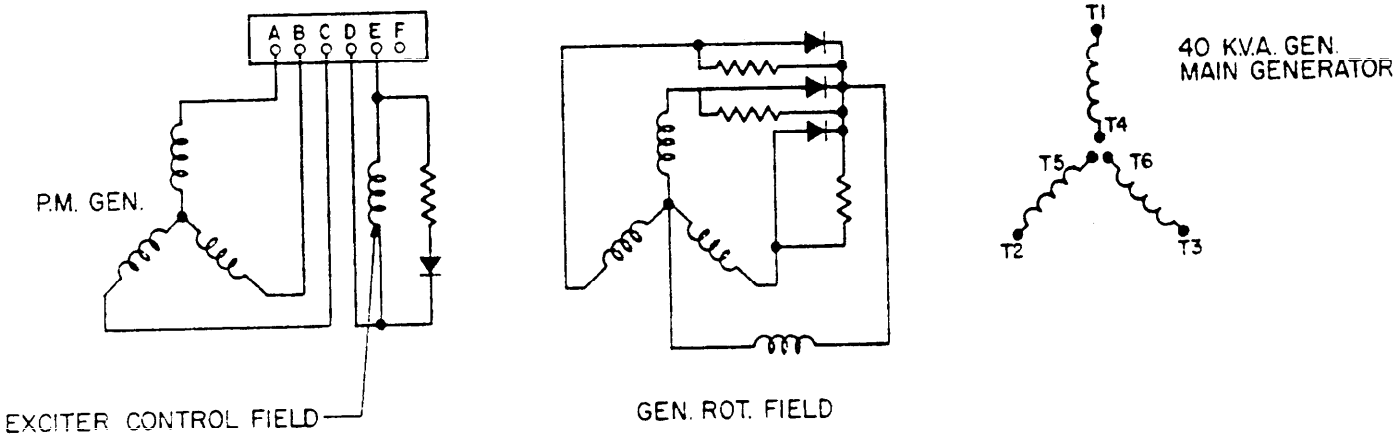
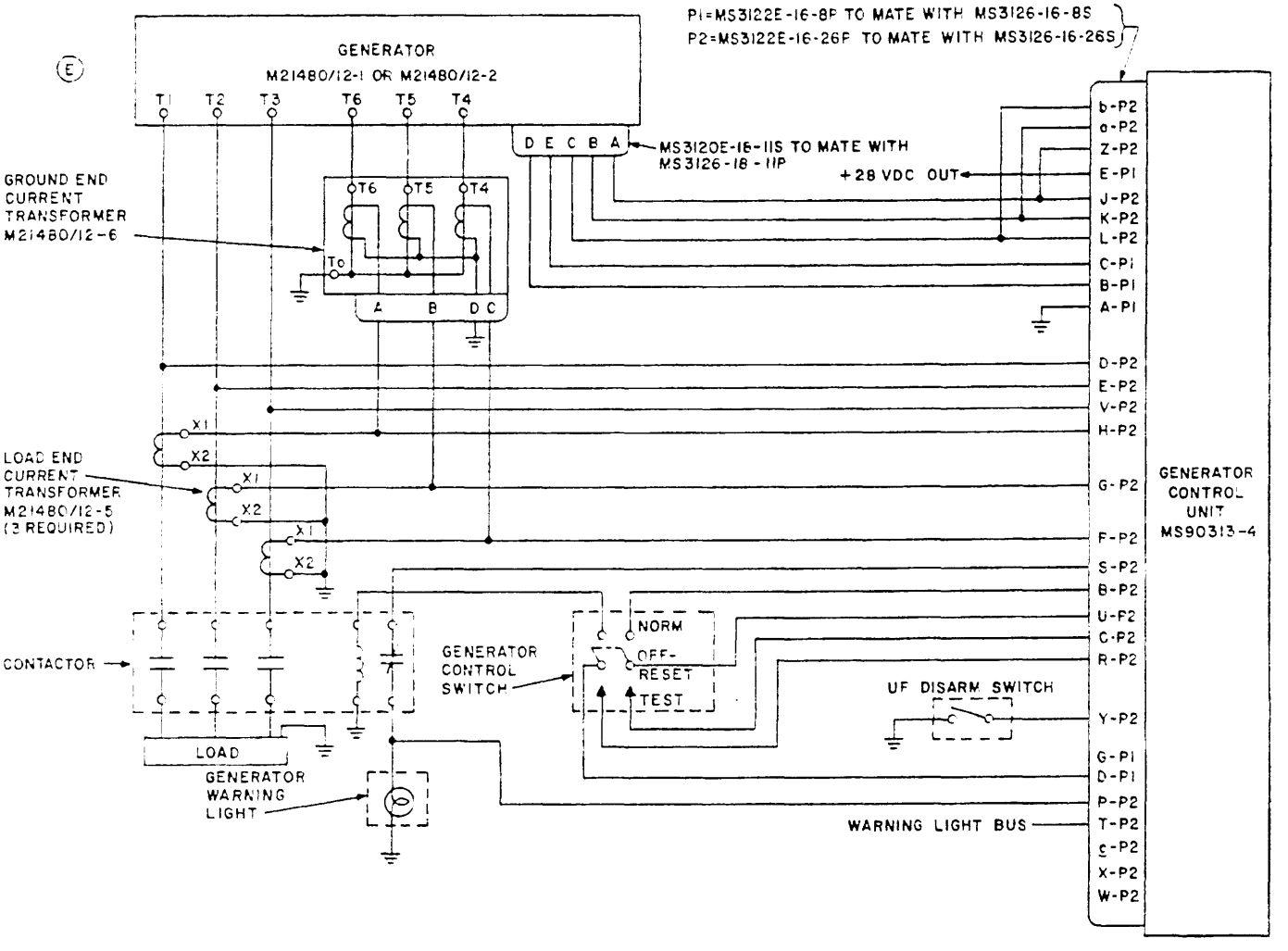


FIGURE 5

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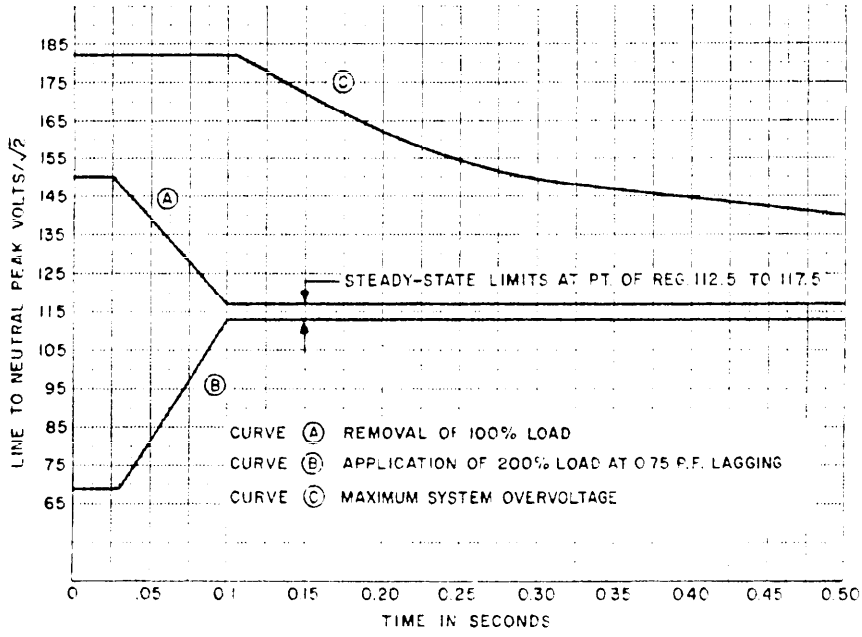
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FIGURE 6 TEST CIRCUIT

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ALTITUDE (Feet)	INLET AIR TEMPERATURE DEGREES CENTIGRADE (°C)	GENERATOR MAXIMUM CONTINUOUS LOAD	
		.75 POWER FACTOR	UNITY POWER FACTOR
0 - Sea Level	40	60	63
0 - Sea Level	20	64	67
0 - Sea Level	10	66	68
0 - Sea Level	0	68	71
0 - Sea Level	-20	72	75
10,000	20	55	57
10,000	10	58	60
10,000	0	60	62
10,000	-20	65	68
15,000	20	50	53
15,000	10	51	52
15,000	0	55	57
15,000	-20	59	61
25,000	20	42	4
25,000	10	43	45
25,000	0	47	49
25,000	-20	52	54

(E) FIGURE 7. GENERATOR ALTITUDE RATING CHART



(E) FIGURE 8. TRANSIENT, STEADY-STATE AND OVERVOLTAGE LIMITS

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REQUIREMENTS:

The requirements of MIL-G-21480 apply with the following exceptions:

Delete the following paragraphs:

3.4.2.2 through 3.4.2.8, 3.4.10, 3.4.10.5, 3.4.10.6, 3.4.13.1, 3.4.14, 3.4.15, 3.4.16, 3.4.16.1 through 3.4.16.5, 3.5.7.5, 4.5.20, 4.5.22.1 through 4.5.22.3, 4.5.22.3.1 through 4.5.22.3.4, 4.5.24.1 through 4.5.24.3, 4.5.24.3.1 through 4.5.24.3.4, 4.5.25.1, 4.5.25.2, 4.5.25.2.1, 4.5.28, 4.5.32, 4.5.33, 4.5.33.1, 4.5.33.2.

3.4.2 Delete and add: Environmental requirements. The system shall meet the requirements of MIL-E-81910 unless otherwise specified by the detailed specification.

(E) 3.4.2.1 Delete and add: Temperature and altitude. The generator shall meet the temperature-altitude requirements of MIL-E-81910 for blast cooled equipment. When operated under self-cooled conditions, the generator shall meet the requirements of Figure 7 of the detailed specification. The control components shall operate continuously when subjected to the temperature-altitude requirements of MIL-E-81910 for self cooled equipment.

(E) 3.4.8 Change "Figure 3" to "Figure 8 of the detailed specification."

3.4.9.3 Add: The generator system shall be designed so that upon removal of any connector plug other than those of the current transformers the system shall be deenergized.

Add 3.4.9.3.1 Wiring schematic. Generator wiring shall be in accordance with Figure 5.

3.4.9.6 Add: The generator shall be self-cooled and in addition shall be provided with means for installation of an air inlet tube.

(E) 3.4.9.6.1.1 Delete and add: Altitude rating chart. The nominal system rating shall be 40 KVA and values of generator maximum continuous loads from sea level to 25,000 feet shall be as shown in Figure 7 of the detailed specification.

3.4.10.3 Change "5000 hours" to "4000 hours."

3.4.11 Change "All system components" to "All system components except current transformers."

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3.4.17 Delete and add: Reliability. The system MTBF (Mean Time Between Failures) shall be not less than 1200 hours. This figure shall apply to failures which will cause system operation to deviate from performance requirement.

(E) 3.5.1 Change "Figure 3" to "Figure 8 of the detailed specification."

3.5.4 Delete and add: Wave form. The crest factor for each phase voltage wave form shall be 1.41 ±10%. No single harmonic shall exceed 3% of the fundamental.

3.5.5 Delete and add: Integral control power. The system shall provide control power which is capable of supplying to the line contactor a continuous output of 3 amperes at 28 volts DC (nominal) and an intermittent output of 10 amperes for 50 milliseconds at a minimum of 20 volts DC. The permanent magnet generator used for integral control power shall recover to normal voltage after a short circuit.

(E) 3.5.7.2 Change "Figure 6" to "Figure 8 of the detailed specification."

3.5.7.3 Delete and add: Undervoltage. Undervoltage protection shall be provided by a static element which functions to disconnect the system from the load bus and deenergize the generator within a period of 6 ± 1 seconds after any phase voltage (line-neutral) falls to 100 ± 5 volts. The system shall be capable of being reset within 2 seconds after the lowest phase voltage exceeds 105 volts.

3.5.7.4 Delete and add: Underfrequency. Underfrequency protection shall be provided by a static element which functions to disconnect the system from the load bus within a period of from 2 to 5 seconds after the frequency falls to 375 ± 5 Hz. After either automatic or manual reset, the underfrequency protection shall provide for the system to be automatically reconnected to the bus within 2 seconds after the frequency exceeds 378 ± 3 Hz. The spread between dropout and pickup shall not be less than 4 Hz.

3.5.7.6 Change "25 amperes" to "18 ± 3 amperes."

3.5.8 Delete and add: Electromagnetic interference. The system shall meet the requirements of MIL-E-81910 Class No. IIIB for the generators and control equipment, except the ten microfarad feedthrough capacitor shall be removed.

4.2.1.1 Add: Qualification tests shall be supplemented with a minimum 100 hour failure free flight test before consideration is given to incorporating the item on the Qualified Products List.

(E) 4.5.3.1 Change "Figure 3" to "Figure 8 of the detailed specification."

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4.5.3.2 Change "Figure 3" to "Figure 8 of the detailed specification."

4.5.12 Delete and add: Electromagnetic interference. The system shall be subjected to an electromagnetic interference test in accordance with MIL-E-81910 for Class IIIB equipment.

4.5.13 Delete and add: Wave form. The crest factor and harmonic content, line-line and line-neutral, of the output voltage shall be determined under the following conditions:

a. At generator speed of 6000 RPM for no load, and in 20% load increments to full rated load, unity power factor.

b. Condition (a) shall be repeated for 0.75 power factor.

c. Conditions (a) and (b) shall be repeated at generator speeds of 5700 and 6300 RPM.

The following requirements must be met during all above tests. The line-neutral crest factor shall be 1.41 ± 10%. No harmonic shall exceed 3% of the fundamental.

4.5.20.1 Delete and add: Salt fog, control components. Control components shall be subjected to a salt fog test in accordance with MIL-E-81910, except the duration of the test shall not exceed 25 hours. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effects of application and removal of rated and 200% load, and protective functions.

4.5.20.2 Delete and add: Oil-salt water ingestion, generators. The generator shall be subjected to an oil-salt water ingestion test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics and the effect of application and removal of rated and 200% load.

4.5.21 Delete and add: Fungus resistance. System components shall be subjected to a fungus test in accordance with MIL-E-81910, with the installation instructions and mounting nuts attached. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions. Mounting nuts are to be unaffected and the package holding the nuts suitable for shipment. Instructions are to be legible.

4.5.22 Delete and add: Sand and dust. The generator shall be subjected to a sand and dust test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics and the effect of application and removal of rated and 200% load.

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205. Varnish missing Visual

4.5.23.1.1 Delete and add: Generator. The generator shall be subjected to a vibration test in accordance with MIL-E-81910, except that the amplitude of vibration monitored at the anti-drive end of the generator during the main bending mode shall be limited to 25g's but the vibration input shall not be lowered below 5g's. At the completion of this test the unit shall successfully demonstrate its buildup characteristics and the effect of application and removal of rated and 200% load.

4.5.23.1.2 Delete and add: Control components. Control components shall be subjected to a vibration test in accordance with MIL-E-81910, except that the vibration range shall be from 5 to 500 Hz. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.24 Delete and add: Shock. System components shall be subjected to a shock test in accordance with MIL-E-81910. At the completion of procedure 1, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions. There shall be no failure of the mounting attachments and the test item shall remain in place and not create a hazard during procedure III.

4.5.25 Delete and add: Humidity. System components, except those tested under paragraph 4.5.20.2, shall be subjected to a humidity test in accordance with MIL-E-81910. At the completion of this test, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.30 Delete first sentence and add: All system components except the generator shall be subjected to the acceleration tests of MIL-E-81910. At the completion of the tests the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

NOTES

1. Components enclosed in dotted lines are shown for interconnections only and are not furnished under this drawing. Connecting leads shown shall be provided by the aircraft manufacturer.

2. Dimensions in inches. Unless otherwise specified, tolerances: linear $\pm$.031, angular $\pm 1^{\circ}$.

3. All outline dimensions are limiting dimensions only.

4. Thread dimensions and designations shall be interpreted in accordance with Handbook H 28 and MIL-STD-9, respectively.

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(E) 5. The difference in the M21480/12-1 and M21480/12-2 generators is that the M21480/12-1 uses Vanadium Permendur steel.

ITEM	PART NUMBER
Complete System (Consists of one Generator, one Generator Control Unit, three Load End Current Transformers, and one Ground End Current Transformer	M21480/12-40
Generator	M21480/12-1
(E) Generator	M21480/12-2
Generator Control Unit	MS90313-4
Load End Current Transformer	M21480/12-5
Ground End Current Transformer	M21480/12-6

For design feature purposes, this standard takes precedence over procurement documents referenced herein. Referenced documents shall be of the issue in effect on date of invitations for bid.

Preparing Activity:
Navy - AS
Project No. 6115-N476

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